Problem Set-4.

Hamming Codes and Boolean algebra.

1] Optional, Ans: bit to be corrected was 6.

2] a) 10110101001.

1 2 3 4 5 6 7 8 9 10 11

$P_1 \rightarrow 1357911 \rightarrow 110001 \rightarrow C_1 = 0$

$P_2 \rightarrow 2367910 \rightarrow 111000 \rightarrow C_2 = 0$

$P_4 \rightarrow 4567 \rightarrow 1010 \rightarrow C_3 = 1$

$P_8 \rightarrow 891011 \rightarrow 1001 \rightarrow C_4 = 1$

→ Bit to be corrected = 12th bit.

Corrected Code: 10010101001

→ The given code sent is correct and doesnot require any change.

(b) 11011101.

1 2 3 4 5 6 7 8

$P_1 \rightarrow 1010 \rightarrow C_1 = 1$

$P_2 \rightarrow 10111 \rightarrow C_2 = 1$

$P_4 \rightarrow 1110 \rightarrow C_3 = 0$

$P_8 \rightarrow 1 \rightarrow C_4 = 0$

bit to be corrected = 3.

Correct word:

11111101.

3] (a)

A	B	$\bar{A}B$	AB	$A+\bar{A}B+AB$	$A+B$
0	0	0	0	0	0
0	1	1	0	1	1
1	0	0	0	1	1
1	1	0	1	1	1

Same.

(b)

A	B	$A+\bar{B}$	$\bar{A}+B$	$(A+\bar{B})(\bar{A}+B)$	AB	$\bar{A}\bar{B}$	$AB+\bar{A}\bar{B}$
0	0	1	1	1	0	1	1
0	1	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	1	1	1	1	1	0	1

4] a) $xy + xy + \bar{x}\bar{y} = xy + \bar{x}\bar{y} = 1$.

b) $\underbrace{xyz + \underbrace{yyz}_{yz} + \bar{y}z}_{yz(x+1).} + \bar{x}y$

$z(y+\bar{y}) + \bar{x}y = z + \bar{x}y$.

5] a) $\bar{B}\bar{C}D + \underbrace{[\overline{B+C+D}]}_{\bar{B}\bar{C}\bar{D}} + \bar{B}\bar{C}\bar{D}E$

$\bar{B}\bar{C}[D+\bar{D}] + \bar{B}\bar{C}\bar{D}E$

$\bar{B}\bar{C}[1+\bar{D}E]. = \bar{B}\bar{C}$

c] $\overline{[\bar{A}\cdot\bar{B}] + [\bar{C}\,\bar{D}][\bar{E}\cdot\bar{F}][\bar{G}\cdot\bar{H}]}$

$= \overline{\bar{A}\cdot\bar{B}\cdot\bar{C}\cdot\bar{D}} + \overline{\bar{E}\cdot\bar{F}\cdot\bar{G}\cdot\bar{H}} \quad = (A+B+C+D)(E+F+G+H).$

d] $\overline{[\bar{A}+B+\bar{C}D][0+\bar{A}BC\bar{D}]}$

$= [\bar{\bar{A}}\cdot\bar{B}\cdot C\cdot\bar{D}] + [A+\bar{B}+\bar{C}+\bar{D}] = \cancel{\bar{A}BCD + \bar{A}\cdot\bar{B}}$

8] a) $\overline{\overline{AB + CD(A\bar{B}+CD)}}$

$\overline{(\overline{AB})\cdot\overline{CD(A\bar{B}+CD)}}$

$\hookrightarrow \overline{\overline{A\bar{B}+CD}} = \overline{(\overline{A\bar{B}})\cdot(\overline{CD})}$

9] (a) $\overline{\overline{(xy+z)(xy+p)}} \Rightarrow \overline{\overline{(\overline{xy}\cdot\bar{z})(\overline{xy}\cdot\bar{p})}}$

$\hookrightarrow$ do it separality

(b) $\overline{\overline{(1+A)(ABC)}} \Rightarrow \overline{\overline{(\overline{1}\cdot\bar{A})\cdot(ABC)}}$

10] a) $x\bar{y} + x + \overline{x+y}$

$\overline{\overline{x\bar{y}}} \Rightarrow \overline{\overline{\bar{x}+y}} + x + (\overline{x+y})$

(b) $\overline{\overline{(1+A)(AC)}} = \overline{(\overline{1+A}) + \overline{AC}} = \overline{(\overline{1+A}) + (\bar{A}+\bar{C})}$

5 b] $\overline{(A\bar{B}C)(\overline{\bar{A}B})} + BC$

$= \overline{[\cancel{A\bar{B}C}]} + \overline{[A\bar{B}][\overline{A\bar{B}}]C} + BC$

$= \overline{BC}$

c] $[Wx + W\bar{y}][x+W] + Wx(\bar{x}+\bar{y})$.

$= Wx + Wx + Wx\bar{y} + W\bar{y} + 0 + Wx\bar{y}$

$= Wx(1) + Wx\bar{y}(1) + W\bar{y}$

$= Wx(1+\bar{y}) + W\bar{y}$

$= W(x+\bar{y})$.

6) a) $ABCD + AB(\overline{CD}) + (\overline{AB})CD$

$AB(CD + \overline{CD}) + \cancel{\overline{AB}}\, CD(AB + \overline{AB})$. $= AB + CD$

b) $(A+\bar{A})(AB + AB\bar{C}) = 1[AB(1+\bar{C})] = AB$.

c) $(A\bar{B} + A\bar{C})(BC + B\bar{C})(ABC) = (A\bar{B} + A\bar{C})(ABC) \cancel{+0}$.

$= 0$

d) $A\bar{B}C + \bar{A}BC + ABC = BC(A+\bar{A}) + A\bar{B}C = C[\underbrace{B + A\bar{B}}]$.

$= C\cancel{[B(\bar{B}+\bar{A}B] = C[B+\bar{A}]}[\cancel{\overline{BA\bar{A}B}}$

$= C[\bar{A} + B]$.

7] a] $\bar{P} + \bar{Q}\bar{R}$

b] $[\bar{P}Q] + [R\bar{S}]$.